

30th November 2025
10:31 p.m.

MAT III

Principles of Mathematics

fall 2025

Take-home mid

Amitrakshar Biswas

10000 58970

Ques ①

A E

$$6 \cdot 5 \cdot 4 \cdot 3 = 360 \text{ possibilities}$$

Ans: 3600

A E
· A E
· · A E
· · · A E
· · · · A E

E A
· E A
· · E A
· · · E A
· · · · E A

Ques ④

1 point $\Rightarrow$ infinitely many hemi-spheres

2 point $\Rightarrow$ worst-case: They lie in a great circle G

$\Rightarrow$ 2 candidate hemi-spheres /w G

3 point $\Rightarrow$ 1 candidate hemi-spheres /w G

4th point $\Rightarrow$ This one could be in the other hemi h_2 sphere.
 $\Rightarrow$ 3 points in hemisphere h_1
3 points in hemisphere h_2

5th point $\Rightarrow$ is in $h_1 \rightarrow$ choose $h_1 \rightarrow$ get 4 in h_1
is in $h_2 \rightarrow$ choose $h_2 \rightarrow$ get 4 in h_2 $\square$

Ques (2)

set A = Begins with 1

$$|A| = 2^6$$

set B = Ends with 1

$$|B| = 2^6$$

C = Begins /w ①
and
Ends /w ①

$$|C| = 2^5$$

D = Begins /w ①
OR
Ends /w ①

$$|D| = |A| + |B| - |C| = 128 - 32 = 96$$

F = exactly four ①s

$$|F| = {}^7C_4 = 35$$

G = " " ①s
and
Begins /w ①

$$\longrightarrow {}^6C_3 = 20$$

H = " " ①s
and
ends /w ①

$$\longrightarrow {}^6C_3 = 20$$

I = exactly four ①s
and
Begins /w ①
and
ends /w ①

$$\longrightarrow {}^5C_2 = 10$$

$$\text{Ans} = D + \{F - (G + H - I)\} = 96 + 5 = 101.$$

Ques 3 $\Gamma : [0, \infty) \rightarrow \mathbb{R}$

$$\Gamma(x) = \int_0^{\infty} t^{x-1} \cdot e^{-t} dt$$

remarkable property,

$$\Gamma(n) = (n-1)! \quad \forall n \in \mathbb{N}$$

Checking for $x = 1, 2, 3, 4$

$$\int_0^{\infty} t^0 \cdot e^{-t} dt = \int_0^{\infty} e^{-t} dt = 1 = 0!$$

$$\int_0^{\infty} t^1 \cdot e^{-t} dt = \int_0^{\infty} t \cdot e^{-t} dt = 1 = 1!$$

$$\int_0^{\infty} t^2 \cdot e^{-t} dt = \int_0^{\infty} t^2 \cdot e^{-t} dt = 2 = 2!$$

$$\int_0^{\infty} t^3 \cdot e^{-t} dt = \int_0^{\infty} t^3 \cdot e^{-t} dt = 6 = 3!$$

Computing $\Gamma(n)!$

$$\Gamma(n)! = \Gamma(\Gamma(n)+1)$$

$$\Gamma(x) = \int_0^{\infty} t^{x-1} \cdot e^{-t} dt$$

$$\Gamma(\Gamma(n)+1) = \int_0^{\infty} t^{\Gamma(n)+1-1} \cdot e^{-t} dt$$

$$= \int_0^{\infty} t^{\Gamma(n)} \cdot e^{-t} dt$$

Ques 5:

- i) Colonel M is not guilty $\Rightarrow$ Crime in Library (a) ^{by given}
- ii) Colonel M is guilty $\Leftarrow$ Crime @ Midnight (c) ^{by given}
- iii) Weapon $\neq$ Wrench $\Leftrightarrow$ Prof. is innocent (d) ^{by given}
- iv) Weapon $\neq$ Wrench $\Leftrightarrow$ Crime in Billiard Room. (b) ^{by given}

a) The Crime took place in The billiard Room
 $\Rightarrow$ iv $\Rightarrow$ iii $\Rightarrow$ Prof = innocent $\Rightarrow$ Mustard = Guilty.

b) The Crime didn't take place in The billiard Room
 $\Rightarrow$ $\neg$ iv $\Rightarrow$ $\neg$ iii $\Rightarrow$ Prof = guilty.

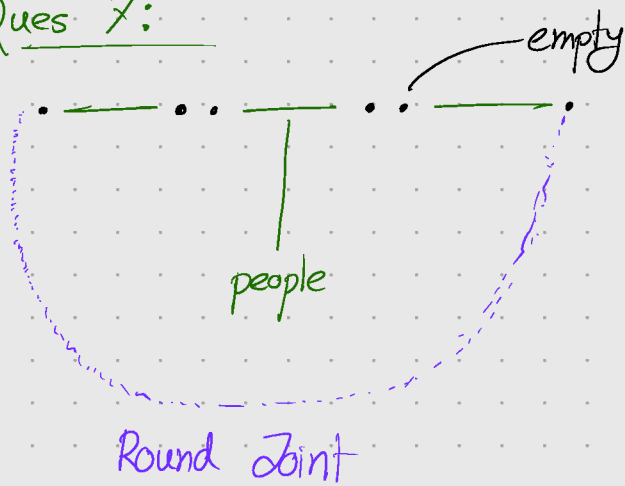
c) The Crime was committed @ Noon with knife
 $\Downarrow$
iv
 $\Downarrow$
iii $\Rightarrow$ Prof = innocent
 $\swarrow$
Colonel can still be guilty

d) The crime took place at midnight in The kitchen
 $\Downarrow$
ii $\Rightarrow$ Colonel = guilty.

Ques 6:

$$24 + 23 + 22 + \dots + 12 = 234$$

Ques 7:



9 chairs	=	3 people minimum seated
12	"	= 4 "
42	"	= 14 "
60	"	= <u>20</u> "

Ans = 20.

Ques 8:

$$\binom{5}{2} \times \binom{9}{2} = 360$$

$$\binom{5}{3} \times \binom{9}{1} = 90$$

$$\binom{5}{4} = 5$$

Ans = 455

Ques 9

No Restrictions $\Rightarrow 16!$

ways

But that includes
"Putting on Shoe before Sock"

So, for each leg $\rightarrow$ gotta divide by 2

8 legs $\rightarrow$ " " " 2^8

$$\text{Ans} = \frac{16!}{2^8}$$

Ques (10)

$$1+2+3+\dots+9 = \frac{10 \times 9}{2} = 45$$

10, 11, 12, ..., 50

1, 2, 3, ..., 41

(negating 9) $\left. \vphantom{\begin{matrix} 10, 11, 12, \dots, 50 \\ 1, 2, 3, \dots, 41 \end{matrix}} \right\} \rightarrow 41 \times 9 = 369$

Ans. $369 + 45 + 1 = 415$

Ques (11)

10 % % %

↓

50%

↓

33%

OR

66%

↓

50%

↓

25%

OR

50%

OR

75%

↓

50%

every individual throw
has 50% prob. of "hitting"

↓

Ans: 50%

for hitting first 50 outta 100

Ques (12)

$$2n+1 = n + (n+1)$$

$$= (n-1) + (n+2)$$

$$= (n-2) + (n+3)$$

⋮

⋮

⋮

$$= 1 + 2n$$

n -lines of "Decompositions"

So, if we take $n+1$ elements from $\{1, 2, 3, \dots, 2n\}$

We will have one of those lines Completed $\square$

Ques 13

Consider, taking n numbers

$$A = \{3n, 3n-2, 3n-4, 3n-6, \dots, n+2\} \quad \text{case 1}$$

$$A = \{3n-1, 3n-3, 3n-5, \dots, n+1\} \quad \text{case 2}$$

- All of Them odd
- None of Them divides each other. — property P

So, our $(n+1)^{\text{th}}$ choice would be a number k

$$n \geq k \quad \text{case 1}$$

$$n-1 \geq k \quad \text{case 2}$$

Case 1

$$k = n \quad 3k = 3n$$

$$k = n-2 \quad 3k = 3n-6$$

$$k = n-4 \quad 3k = 3n-12$$

⋮

$$k = 5 \quad \text{divides } 1+ \text{ number in } A$$

$$k = 3 \quad \text{divides } 1+ \text{ number in } A$$

So, for any choice of k as $(n+1)^{\text{th}}$ number
we would have to remove 1 or more elems from A .

- So, if we want to maintain The property P

The only possible operation is Swapping.

- But, by Swapping, A could never have $> n$ elements.

$\Rightarrow$ as soon as $(n+1)$ element P is broken $\square$

Some way case 2 can be Proven!

Ques 14

The Ques. can be transformed into a Graph Problem. That is,

Undirected Graph

Nodes = People

Edges = Acquaintance Relation

No matter How many edges we take in whatever way

Property P: There will always be 2 nodes having the same # of edges

Say we have n nodes

$\text{degree}(\text{node}) \in \{0, 1, 2, 3, \dots, n-1\}$ total n possible values

n nodes

n distinct degrees

$\Rightarrow$ set of degrees = $\{0, 1, 2, 3, \dots, n-1\}$

Contradiction

$n-1 \Rightarrow$ a node that's connected to all other node

$\Rightarrow$ so no node can have $\text{deg} = 0$

Thus Property P is Proven by Contradiction

& with that, so is the Original Ques.

Ques 15

$n \times n$ Matrix $\Rightarrow 2n+2$ sums

Anti-Magic Square req. $2n+2$ req distinct sums

min. sum $-n$

max. sum n

$\Rightarrow$ possible sums $= \{-n, -n+1, -n+2, \dots, 0, 1, 2, 3, \dots, n\}$

Total possibilities. $= 2n+1 < \underline{2n+2}_{\text{req.}}$

$\therefore$ by Contradiction,

Anti-Magic Square is not Possible.

Ques 16

Well $\sum_{i=0}^{n-1} P_i = 1 = \text{req fuel to take a round once.}$

So obviously yes, we could.

Ques 17

a) $\underbrace{|-2+5|}_3 \neq \underbrace{|-2|+|5|}_7$ $\begin{matrix} x = -2 \\ y = 5 \end{matrix}$

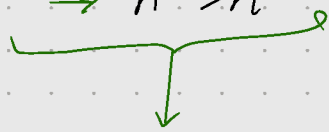
b) $\left. \begin{array}{l} \text{if } 12 \mid 4 \times 3 \\ \text{then } 12 \mid 4 \text{ OR } 12 \mid 3 \end{array} \right\} \text{This implication isn't TRUE}$

c) $\underbrace{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}}_A > 2$ $\underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12}}_B > 1$
 $A + B > 3$

Ques 18

$$n > 1 \Rightarrow n^2 > 3 \Rightarrow n^2 > n$$

$$n < 1 \Rightarrow n^2 > 3 \Rightarrow n^2 > n$$



$$n^2 \mid n$$

big | small

$$\begin{cases} n^2 = n \times n \\ n^2 = n \times n \end{cases}$$

is false $\Rightarrow$ Contradiction

$$n = 1 \quad n^2 = 1$$

$$n = 0 \quad n^2 = 0$$

$$n = -1 \quad n^2 = 1$$

$$n^2 \mid n \quad \text{TRUE}$$

$$n^2 \mid n \quad \text{TRUE}$$

$$n^2 \mid n \quad \text{TRUE}$$



Ques 19

Consider The negation

$$\frac{x+y}{2} < \sqrt{xy}$$

$$\text{let } x = 1$$

$$y = 3$$

$$\frac{4}{2} < \sqrt{3}$$

$$2 < \sqrt{3}$$

false $\Rightarrow$ Contradiction

by Contradiction,

The opposite $\rightarrow$

$$\frac{x+y}{2} \geq \sqrt{xy}$$

must be TRUE.

Ques 20

Say, $\gcd(a, b+am) > 1$

= 2 OR 3 OR bigger

= say $(k) > 1$ $k \in \mathbb{Z}$

$$\textcircled{iii} \quad k \mid a \quad \Rightarrow \quad k \mid am \quad \textcircled{i}$$

$$k \mid b + am \quad \textcircled{ii}$$

$$\textcircled{i} + \textcircled{ii} \Rightarrow k \mid b \quad \textcircled{iv}$$

$$\textcircled{iii} + \textcircled{iv} \Rightarrow \gcd(a, b) = k$$

but $k > 1$ $k \in \mathbb{Z}$

and

ques said $\gcd(a, b) = 1$

Contradiction

$\gcd(a, b+am) = 1$ must be TRUE

Ques (21)

let's consider the **Negation**

$$\gcd\left(\frac{m}{d}, \frac{n}{d}\right) = k > 1 \quad \in \mathbb{Z}$$

$$\gcd(m, n) = kd$$

Contradiction

$$\begin{array}{ll} \text{let } m = dx_0 & x_0 = m/d \\ n = dy_0 & y_0 = n/d \end{array} \quad \gcd(x_0, y_0) = 1$$

$\therefore x_0 \wedge y_0$ are **co-primes**.

Ques (22)

non-negative linear combination $\Rightarrow k_1 a + k_2 b$

$$\text{let, } ab - a - b = k_1 a + k_2 b$$

$$ab = (k_1 + 1)a + (k_2 + 1)b$$

$$\Rightarrow a \mid (k_2 + 1)b$$

$$b \mid (k_1 + 1)a$$

$$\gcd(a, b) = 1$$